

Suvojit Baidya

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Education

Khulna University of Engineering & Technology (KUET) - B.Sc. in Mechanical Engineering Jan 2020 – Oct 2025

Research Interests

Additive Manufacturing	Finite Element Analysis	Sandwich Structures
Origami Structures	Lightweight Structures	Composite Materials
Harmonic Analysis	Bioinspired Structures	Topology Optimization

Publications and Research Experience

Published Article

- Experimental and Numerical Study on Flyash-Reinforced Core of 3D-Printed PETG/WPLA Honeycomb Sandwich Structure
 - * Simulated and validated force displacement and MOE of the sandwich structures
 - * Analyzed energy absorption variation and failure modes with and without flyash-reinforcements

Under Review

- Effect of Material Orientation and Core Geometry on Bending Behavior of PLA+ and ABS-Based 3D-Printed Honeycomb Sandwich Structure
 - *Optimized EA of honeycomb sandwich structure for various cell thickness
 - *Numerically studied various cell thickness in flexural test of honeycomb structures
- Flexural Behavior and Specific Energy Absorption of Octopus Suction-Cup-Inspired Additively Manufactured Sandwich Cores: An Experimental and Numerical Study.
 - *Generated bio-inspired cores with tunable cavity diameter and shell thickness
 - *Measured SEA, plateau stress, and failure modes under quasi-static compression and bending; validated with Ansys
 - *Performing geometry optimization to balance stiffness-to-weight with energy absorption.

Undergraduate Thesis

- Experimental and Numerical Investigation on Effect of Radius of Curvature of Miura-Ori inspired Origami Structure.
 - Miura-Ori inspired parallelogram shaped structure's quasi-static and cyclic compressive behaviour with DMA
 - *Modeled curved-crease Miura-Ori panels with varied crease radius
 - *Analyzing Quasi-static and cyclic compression behavior focusing on SEA, hysteresis area and recoverability

Leadership and Co-Curricular Activities

Captain, Team Kilo Flight (Formula SAE), KUET 2024 – 2025

- Directed cross-functional subteams (chassis, aero, powertrain) from concept to design reviews and build planning.

Body Chassis and Aerodynamics Executive, Team Kilo Flight (KUET) 2023 – 2024

- Developed and designed jute composite body for FSAE vehicle and aerodynamics performance.

Project Experiences

Design & Development of Formula SAE Electric Vehicle

Team Captain

- Led first EV design of Team Kilo Flight; developed chassis and composite with structured assembly and testing.
- Applied CFD and FEA to guide bodywork and structural decisions; prepared manufacturing drawings and BOM.

Awards & Achievements

Formula SAE Japan (2023) (30th of 77 teams)

Cleared Mechanical Inspection as a First Year Vehicle: first team from Bangladesh

- **Special Recognition Award for Co-curricular Excellence** - Director of Student Welfare, KUET
- **Technical Board Scholarship:** Outstanding Higher Secondary Certificate results.

Certificates & Badges

ANSYS Learning Pathways (competency-based)

- *Harmonic Analysis:* Modal/Harmonic response, damping, strain & deformation analysis, fatigue-aware design checks.
- *Optimization:* Topology optimization and design-space constraints for lightweight structures.
- *Composites & FSAE:* Chassis analysis and composite monocoque workflows (laminate layup, failure criteria).
- *Foundation in Stress Analysis in Structures:* Stress in Local Equilibrium, Strain in Deformation, Thermal Strain.

Workshops

- CFD Simulation Workshop — IMechE, KUET (foundation to publication-oriented practice).

Technical & Software Skills

Simulation/FEA: ANSYS (Fluent/CFD, Mechanical: Structural, Rigid & Explicit Dynamics), ABAQUS

CAD & Rendering: SolidWorks, AutoCAD, KeyShot

Additive Manufacturing: Bambu Studio

Programming: C, Python

Hands-on Experience

- **Machining:** drilling, CNC milling, grinding, and metal cutting for prototype brackets & fixtures; tolerance checks with calipers/micrometers.
- **Additive Manufacturing:** FDM printing (Bambu Studio); part orientation/support strategies; post-processing for fit & finish.
- **Test & Measurement:** UTM operation for flexural/compressive tests; specimen preparation and data logging for reports.

Language Proficiency

English: Professional proficiency

IELTS: 7.5 overall, no band less than 7